



The k -general d -position problem for graphs[☆]

Brent Cody^{*}, Garrett Moore

Virginia Commonwealth University, Department of Mathematics and Applied Mathematics, 1015 Floyd Avenue, PO Box 842014, Richmond, VA, 23284, United States

ARTICLE INFO

Article history:

Received 9 September 2024

Received in revised form 7 January 2025

Accepted 13 January 2025

Available online 24 January 2025

MSC:

05C69

05C35

05C38

05C12

Keywords:

General position

Maximally even sets

Cycles

Paths

Grids

Isometric subgraphs

ABSTRACT

A set of vertices of a graph is said to be in general position if no three vertices from the set lie on a common shortest path. Recently Klavžar, Rall and Yero generalized this notion by defining a set of vertices to be in general d -position if no three vertices from the set lie on a common shortest path of length at most d . We generalize this notion further by defining a set of vertices to be in k -general d -position if no k vertices of the set lie on a common shortest path of length at most d . The k -general d -position number of a graph is the largest cardinality of a k -general d -position set. We provide upper and lower bounds on the k -general d -position number of graphs in terms of the k -general d -position number of certain kinds of subgraphs. We compute the k -general d -position number of finite paths and cycles. Along the way we establish that the maximally even subsets of cycles, which were introduced in Clough and Douthett's work on music theory, provide the largest possible k -general d -position sets in n -cycles. We generalize Klavžar and Manuel's notion of monotone-geodesic labeling to that of k -monotone-geodesic labeling in order to calculate the k -general d -position number of the infinite two-dimensional grid. We also prove a formula for the k -general d -position number of certain thin finite grids, providing a partial answer to a question asked by Klavžar, Rall and Yero.

Published by Elsevier B.V.

1. Introduction

The classical no-three-in-line problem, posed by Dudeney [13] over one-hundred years ago asks: What is the largest number of points that can be placed in the $n \times n$ grid so that no three points lie on a line in the plane? Despite being widely studied for many years [16,17,19,31,32], this problem remains open. A graph-theoretic variation of the no-three-in-line problem called the *general position problem for graphs*, introduced independently by Klavžar and Manuel [28] and by Chandran and Parthasarathy [6], asks: Given a graph G , what is the largest number of vertices that can be selected so that no three lie in a common shortest path?

This problem for graphs has itself been extensively studied by many authors [1,18,20,23–25,28,29,36]. A generalization of the classical no-three-in-line problem, recently studied in [27], can be obtained by replacing “three” with an arbitrary integer parameter: What is the largest number of points that can be placed in the $n \times n$ grid so that no k points lie on the same line? Such variations for arbitrary point sets in $\mathbb{R}^2$ have also been studied by Erdős [14,15] and many

[☆] This work was partially funded by Virginia Commonwealth University, United States, through the College of Humanities and Sciences Seed Grant Program.

^{*} Corresponding author.

E-mail addresses: bmccody@vcu.edu (B. Cody), mooregk@vcu.edu (G. Moore).

URL: <http://www.people.vcu.edu/~bmccody/> (B. Cody).

others [2,21,30,34]. In this article, we introduce a variation of the general position problem for graphs in which sets of vertices may have three vertices in a common shortest path, but some larger number of vertices being in a common shortest path is forbidden. Before we formally state the problem, let us review some definitions.

Suppose $G = (V(G), E(G))$ is a graph. The *length* of a path P of G , written as $\lambda_G(P)$, or as $\lambda(P)$ when the context is clear, is the number of edges in P . For vertices $u, v \in V(G)$, the length of a shortest path from u to v in G is denoted by $d_G(u, v)$, or just $d(u, v)$ when the context is clear. A path P in G is called a *geodesic* of G or a *shortest path* of G if it is a shortest path between two vertices of G . The set of vertices corresponding to some geodesic P of G will be denoted by $V(P)$. Notice that for any path P of G , $|V(P)| = \lambda(P) + 1$. We let $\mathcal{G}(G)$ denote the set of all geodesics of G .

Using the terminology of Klavžar and Manuel [28], a set of vertices $S \subseteq V(G)$ is said to be a *general position set* in G provided that

$$\text{for all } P \in \mathcal{G}(G) \text{ we have } |S \cap V(P)| < 3;$$

that is, no three vertices in S lie in a common geodesic. The *general position problem* for G is to find a largest general position set of G . The *general position number* of G , written as $\text{gp}(G)$, is the largest cardinality of a general position set.

Klavžar, Rall and Yero [22] generalized the notion of general position set as follows. For an integer $d \geq 1$, we say that a set of vertices $S \subseteq V(G)$ is a *general d -position set* in G provided that

$$\text{for all } P \in \mathcal{G}(G), \text{ if } |S \cap V(P)| \geq 3 \text{ then } \lambda(P) > d.$$

The *general d -position problem* for G is to find a largest general d -position set in G , and the *general d -position number* of G , written as $\text{gp}_d(G)$, is the largest cardinality of a general d -position set. In the current article, we study a further generalization of general position sets.

Definition 1.1. For integers d and k with $d \geq 1$ and $k \geq 2$, we say that a set of vertices $S \subseteq V(G)$ is a *k -general d -position set* in G , or is in *k -general d -position* in G (where *in G* may be omitted when the context is clear) provided that

$$\text{for all } P \in \mathcal{G}(G), \text{ if } |S \cap V(P)| \geq k \text{ then } \lambda(P) > d. \quad (1.1)$$

The *k -general d -position problem* for G is to find a largest k -general d -position set in G . The *k -general d -position number* of G , denoted by $\text{gp}_d^k(G)$, is the largest cardinality of a k -general d -position set in G .

A few easy observations are in order. Recall that a set of vertices of a graph is *independent* if no edge of the graph connects two vertices of the set. The *independence number* of G , written as $\alpha(G)$, is the largest size of an independent set in G . It is easy to see that a set of vertices A is a 2-general 1-position set in G if and only if it is independent. Thus $\text{gp}_1^2(G) = \alpha(G)$, and since a set being in 2-general d -position, for some $d \geq 1$, implies that it is in 2-general 1-position, it follows that 2-general d -position sets can be thought of as having a strong form of independence. Indeed, a set of vertices is in 2-general d -position if and only if it is *d -distance independent* in the terminology of [33]. Hence, the *d -distance independence number* $\alpha_d(G)$ [33], which is the largest size of a d -distance independent set, satisfies

$$\text{gp}_d^2(G) = \alpha_d(G).$$

For $d \geq 1$, since a set of vertices S is a 3-general d -position set in G if and only if it is a general d -position set, we have

$$\text{gp}_d^3(G) = \text{gp}_d(G).$$

Furthermore, for $k \geq 2$ and $d \geq \text{diam}(G)$, a set of vertices $S \subseteq V(G)$ is a k -general d -position set in G if and only if

$$\text{for all } P \in \mathcal{G}(G) \text{ we have } |S \cap V(P)| < k \quad (1.2)$$

(note the independence of this notion from d). Thus,

$$\text{if } d \geq \text{diam}(G) \text{ then } \text{gp}_d^k(G) = \text{gp}_{\text{diam}(G)}^k(G),$$

and we will often write $\text{gp}^k(G)$ to mean $\text{gp}_{\text{diam}(G)}^k(G)$ in this case to emphasize the independence from d . Indeed, a set $S \subseteq V(G)$ is said to be a *k -general position set* provided that it is a k -general $\text{diam}(G)$ -position set in G , i.e., (1.2) holds.

Notice that if G has at least $k - 1$ vertices then $\text{gp}^k(G) \geq k - 1$. For $d \geq 1$, $k \geq 2$ and $n \geq 1$ we have

$$\text{gp}_d^k(K_n) = \begin{cases} 1 & \text{if } k = 2 \\ n & \text{if } k \geq 3, \end{cases}$$

where K_n is the complete graph on n vertices.

In Section 2, we discuss some general properties of k -general d -position sets, some of which will be used throughout the paper. We establish an ordering that holds between the values of $\text{gp}_d^k(G)$ in general (see Proposition 2.2 and Fig. 1). We prove some results on k -general d -position sets in infinite graphs (see Corollary 2.3, Example 2.4 and Proposition 2.5). In Lemma 2.7, we generalize Klavžar and Manuel's *isometric cover lemma* [28, Theorem 3.1] from general position sets to k -general d -position sets to obtain an upper bound on $\text{gp}_d^k(G)$ in terms of the k -general d -position numbers of certain subgraphs. In Lemma 2.8, we generalize a result of Klavžar, Rall and Yero [22, Proposition 1.1] to obtain an upper bound on $\text{gp}_d^k(G)$ in terms of the k -general d -position number of certain subgraphs of G .

Proposition 2.2. Suppose G is a graph. Let $1 \leq d, d' \leq \text{diam}(G)$ and $k, k' \geq 2$ be integers such that $d' \leq d$ and $k' \geq k$. Then $\text{gp}_d^k(G) \leq \text{gp}_{d'}^{k'}(G)$ (see Fig. 1).

Klavžar, Rall and Yero [22, Proposition 6.1] proved that if G is an infinite graph and $1 \leq d < \infty$, then $\text{gp}_d^3(G) = \infty$. Hence, we obtain the following corollary of [22, Proposition 6.1] and Proposition 2.2, which implies that the problem of determining $\text{gp}_d^k(G)$ is trivial for infinite graphs when $k \geq 3$.

Corollary 2.3. Suppose G is an infinite graph and let $d \geq 2$ and $k \geq 3$ be integers. Then $\text{gp}_d^k(G) = \infty$.

However, when $k = 2$, the quantity $\text{gp}_d^2(G)$ is not infinite for all infinite graphs. For example, when K_κ is an infinite complete graph with cardinality $|V(G)| = \kappa$ we have $\text{gp}_d^2(K_\kappa) = 1$. Evidently, as demonstrated in the next example, the 2-general d -position number of infinite graphs with cardinality κ can take on many different values.

Example 2.4. Let G be the graph obtained by connecting one of the endpoints of P_n with one of the vertices v of K_κ . Since $\text{gp}_d^2(P_n) = \lceil \frac{n}{d+1} \rceil$ (this follows directly from Theorem 3.3), and since we can essentially treat v and $V(K_\kappa) \setminus \{v\}$ as being two additional vertices, we have

$$\text{gp}_d^2(G) = \text{gp}_d^2(P_{n+2}) = \left\lceil \frac{n+2}{d+1} \right\rceil.$$

However, for some infinite graphs, the problem of finding the 2-general d -position number does trivialize, as indicated in the next – self-evident – proposition.

Proposition 2.5. Suppose that either the diameter of G is infinite or G has infinitely many connected components. Then $\text{gp}_d^2(G) = \infty$ for all $d \geq 1$.

Let us now generalize Klavžar and Manuel's *isometric cover lemma* [28, Theorem 3.1] from the general position number to the k -general d -position number. A subgraph H of a graph G is a *convex subgraph* of G if it includes every shortest path in G between two of its vertices. A subgraph H of G is said to be an *isometric subgraph* of G if $d_H(u, v) = d_G(u, v)$ for all $u, v \in V(H)$. Clearly, every convex subgraph of a graph G is an isometric subgraph.

Lemma 2.6. Suppose G is a connected graph and H is a subgraph of G . Let $d \geq 1$ and $k \geq 2$ be integers.

- (1) If H is an isometric subgraph of G then whenever $A \subseteq V(H)$ is a k -general d -position set in G , it follows that A is a k -general d -position set in H .
- (2) If H is a convex subgraph of G then for all $A \subseteq V(H)$, A is a k -general d -position set in G if and only if it is a k -general d -position set in H .

Proof. Suppose H is an isometric subgraph of G and $A \subseteq V(H)$ is a k -general d -position set in G . Suppose $P \in \mathcal{G}(H)$ is a geodesic of H with $|A \cap P| \geq k$. Since H is an isometric subgraph of G , we see that $P \in \mathcal{G}(G)$ and furthermore, since A is a k -general d -position set in G we have $\lambda_G(P) > d$. Again, applying the fact that H is an isometric subgraph of G , we have $\lambda_H(P) = \lambda_G(P) > d$, as desired.

Suppose H is a convex subgraph of G and $A \subseteq V(H)$. Since H is an isometric subgraph of G , it follows from (1) that if A is a k -general d -position set in G then it is a k -general d -position set in H . For the converse, suppose A is a k -general d -position set in H . Let $P \in \mathcal{G}(G)$ be a geodesic of G with $|A \cap P| \geq k$. Let $P = \{v_1, v_2, \dots, v_\ell\}$ where $v_1 v_2 \dots v_\ell$ is a shortest path in G . Let a be the least index of an element of A in P and let b be the greatest index of an element of A in P . Then $v_a v_{a+1} \dots v_b$ is a shortest path in G between two elements of H . Since H is a convex subgraph of G , $v_a v_{a+1} \dots v_b$ is a shortest path in H and so $P' = \{v_a, \dots, v_b\}$ is a geodesic of H such that $|A \cap P'| \geq k$. Since A is a k -general d -position set in H we have $\lambda_H(P') > d$, and since H is an isometric subgraph of G we have $\lambda_G(P) \geq \lambda_G(P') = \lambda_H(P') > d$. So, A is a k -general d -position set in G . $\square$

Generalizing [28, Theorem 3.1], we obtain the following.

Lemma 2.7. Suppose G is a finite connected graph and $\{H_1, \dots, H_\ell\}$ is a collection of isometric subgraphs of G such that $V(G) = \bigcup_{i=1}^\ell V(H_i)$. Suppose d and k are integers with $n \geq 1$ and $k \geq 2$. Then

$$\text{gp}_d^k(G) \leq \sum_{i=1}^\ell \text{gp}_d^k(H_i).$$

Proof. Suppose A is a k -general d -position set in G . By Lemma 2.6 (1), for all $i \in \{1, \dots, \ell\}$, $A \cap V(H_i)$ is a k -general d -position set in H_i and thus $|A \cap V(H_i)| \leq \text{gp}_d^k(H_i)$. This implies that

$$|A| \leq \sum_{i=1}^\ell \text{gp}_d^k(H_i),$$

as desired. $\square$

Next, we generalize a result on general d -position sets due to Klavžar, Rall and Yero [22, Proposition 1.1] to obtain a lower bound on $\text{gp}_d^k(G)$ in terms of isometric subgraphs. Given a graph G with subgraphs H and K , we let

$$d_G(H, K) = \min\{d_G(u, v) : u \in V(H) \text{ and } v \in V(K)\}.$$

Lemma 2.8. Suppose $d \geq 1$ and $k \geq 2$ are integers. Let G be a finite connected graph and let $\{H_1, \dots, H_\ell\}$ be a finite collection of isometric subgraphs of G such that

- (1) if $k = 2$ then $d_G(H_i, H_j) > d$ for $i \neq j$ and
- (2) if $k \geq 3$ then $d_G(H_i, H_j) \geq d$ for $i \neq j$.

Then

$$\sum_{i=1}^{\ell} \text{gp}_d^k(H_i) \leq \text{gp}_d^k(G).$$

Proof. For each $i \in \{1, \dots, \ell\}$, let $S_i \subseteq V(H_i)$ be a k -general d -position set in H_i with $|S_i| = \text{gp}_d^k(H_i)$. Let $S = \bigcup_{i=1}^{\ell} S_i$. Since the vertex sets of the subgraphs in $\{H_1, \dots, H_\ell\}$ are pairwise disjoint, we have

$$|S| = \sum_{i=1}^{\ell} \text{gp}_d^k(H_i).$$

Let us show that S is a k -general d -position set in G . Suppose not, then there is some geodesic P of G such that $|S \cap V(P)| \geq k$ and $\lambda_G(P) \leq d$. By assumption (1) and (2), it follows that there is an $i_0 \in \{1, \dots, \ell\}$ such that $S \cap V(P) \subseteq V(H_{i_0})$, and hence $S \cap V(P) = S_{i_0}$. But, since H_{i_0} is an isometric subgraph of G , it follows that there is a geodesic P' of H_{i_0} such that $V(P') = V(P) \cap V(H_{i_0})$ and $\lambda_{H_{i_0}}(P') \leq \lambda_G(P)$. Thus $|S_{i_0} \cap V(P')| \geq k$ and $\lambda_{H_{i_0}}(P') \leq d$, which contradicts the fact that S_{i_0} is a k -general d -position set in H_{i_0} . $\square$

3. Paths

In the current section we will generalize the following result of Klavžar, Rall and Yero to k -general d -position sets.

Proposition 3.1 (Klavžar, Rall and Yero, [22, Proposition 3.1]). If $n \geq 3$ and $2 \leq d \leq n - 1$, then

$$\text{gp}_d(P_n) = \begin{cases} 2 \left\lceil \frac{n}{d+1} \right\rceil - 1 & \text{if } n \equiv 1 \pmod{d+1} \\ 2 \left\lceil \frac{n}{d+1} \right\rceil & \text{otherwise.} \end{cases}$$

Let us first establish a formula for $\text{gp}_d^k(P_n)$ under the assumption that $n \leq d + 1$. Let us note that we will take the vertices of P_n to be $V(P_n) = \{1, \dots, n\}$ with adjacencies given by $\{i, i + 1\} \in E(P_n)$ for $i \in \{1, \dots, n - 1\}$.

Lemma 3.2. Suppose d, k and n are integers such that $d \geq 1$ and $2 \leq k \leq n \leq d + 1$. Then

$$\text{gp}_d^k(P_n) = \text{gp}^k(P_n) = \min(n, k - 1).$$

Proof. Since $n \leq d + 1$, it follows that there are no geodesics in P_n with length greater than d . Thus, $\text{gp}_d^k(P_n) = \text{gp}^k(P_n)$. It easily follows that

$$A = \{1, \dots, \min(n, k - 1)\}$$

is a k -general position set in P_n , and any set of vertices of P_n with cardinality greater than $\min(n, k - 1)$ is not a k -general position set in P_n . Thus, $\text{gp}_d^k(P_n) = \text{gp}^k(P_n) = \min(n, k - 1)$. $\square$

Now we will prove the general version of the formula for $\text{gp}_d^k(P_n)$.

Theorem 3.3. Suppose d, k and n are integers such that $d \geq 1$ and $2 \leq k \leq n$. Then

$$\text{gp}_d^k(P_n) = \begin{cases} n & \text{if } 1 \leq d \leq k - 2 \\ (k - 1) \left\lfloor \frac{n}{d+1} \right\rfloor + \min(n \bmod (d + 1), k - 1) & \text{if } k - 1 \leq d. \end{cases}$$

Proof. Fix d, k and n as in the hypothesis. The first case of the formula follows easily from Lemma 2.1.

Suppose $k - 1 \leq d$. We define a particular subset $S_{k,d}$ of $\mathbb{Z}_{\geq 1}$ that will be used to define a k -general d -position subset of P_n . Let

$$S_{k,d} = \bigcup_{i=0}^{\infty} [(d + 1)i + 1, (d + 1)i + k - 1].$$

Table 1
The quantities $\text{gp}_d^k(P_{14})$ for various values of k and d .

Values of d	Values of k													
	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	7	14	14	14	14	14	14	14	14	14	14	14	14	14
2	5	10	14	14	14	14	14	14	14	14	14	14	14	14
3	4	8	11	14	14	14	14	14	14	14	14	14	14	14
4	3	6	9	12	14	14	14	14	14	14	14	14	14	14
5	3	6	8	10	12	14	14	14	14	14	14	14	14	14
6	2	4	6	8	10	12	14	14	14	14	14	14	14	14
7	2	4	6	8	10	12	13	14	14	14	14	14	14	14
8	2	4	6	8	10	11	12	13	14	14	14	14	14	14
9	2	4	6	8	9	10	11	12	13	14	14	14	14	14
10	2	4	6	7	8	9	10	11	12	13	14	14	14	14
11	2	4	5	6	7	8	9	10	11	12	13	14	14	14
12	2	3	4	5	6	7	8	9	10	11	12	13	14	14
13	1	2	3	4	5	6	7	8	9	10	11	12	13	14

Define $q = \lfloor \frac{n}{d+1} \rfloor$ and let $r = n \bmod (d+1)$; in other words, let r be the unique integer such that $n = q(d+1) + r$ and $0 \leq r < d+1$. Let $m = \min(n \bmod (d+1), k-1)$. It is easy to verify that

$$|S_{k,d} \cap [1, n]| = (k-1)q + m.$$

The induced subgraph $P_n[S_{k,d} \cap [1, n]]$ has either q or $q+1$ connected components, each of size at most $k-1$. Furthermore, the distance between the least elements of two distinct connected components of $P_n[S_{k,d} \cap [1, n]]$ is at least $d+1$. Thus, if a geodesic $P \in \mathcal{G}(P_n)$ contains k or more vertices of $S_{k,d} \cap [1, n]$, then $\lambda_{P_n}(P) > d$. This implies that $S_{k,d} \cap [1, n]$ is a k -general d -position set in P_n and hence $\text{gp}_d^k(P_n) \geq (k-1)q + m$.

Let us show that $\text{gp}_d^k(P_n) \leq (k-1)q + m$.

Suppose $n < d+1$. Then $q = 0$ and $n \bmod (d+1) = n$. Thus, it follows by Lemma 3.2, that

$$\text{gp}_d^k(P_n) = \min(n, k-1) = (k-1)q + m.$$

Suppose $n = d+1$, then $q = 1$, $r = n \bmod (d+1) = m = 0$, and it follows by Lemma 3.2, that

$$\text{gp}_d^k(P_n) = \text{gp}_d^k(P_{d+1}) = \min(d+1, k-1) = k-1 = (k-1)q + m.$$

Now suppose that $n > d+1$. Then $q \geq 1$. We define a sequence $H_0, \dots, H_q$ of subgraphs of P_n as follows. For $i \in \{0, \dots, q-1\}$, let

$$X_i = [(d+1)i + 1, (d+1)(i+1)].$$

If $r > 0$ let $X_q = [(d+1)q + 1, n]$ and otherwise, let $X_q = \emptyset$. For $i \in \{0, \dots, q\}$ let $H_i = P_n[X_i]$ and note that H_i is an isometric subgraph of P_n . Furthermore, $V(P_n) = \bigcup_{i=0}^q V(H_i)$. Hence by applying Lemma 2.7, and using that $H_i \cong P_{d+1}$ for $i \in \{0, \dots, q-1\}$ and $H_q \cong P_r$ where $r < d+1$, it follows from Lemma 3.2 that

$$\text{gp}_d^k(P_n) \leq \sum_{i=0}^q \text{gp}_d^k(H_i) = \text{gp}_d^k(P_{d+1})q + \text{gp}_d^k(P_r) = (k-1)q + m. \quad \square$$

Example 3.4. The values of $\text{gp}_d^k(P_{14})$ are indicated in Table 1. Note that the values in the $k=3$ column match those of $\text{gp}_d(P_{14})$, which appear in [22, Table 1]. Thus, we see that the values of gp_d , for P_{14} and more broadly for any graph, are a one dimensional slice of a multidimensional phenomenon.

4. Cycles

4.1. Preliminaries on maximal evenness

Clough and Douthett [7] showed that sets of vertices in cycles graphs that are distributed as evenly as possible tend to produce interesting sets of pitch classes in music theory; they refer to such sets as maximally even sets. Later Toussaint [37] showed that the same sets manifest in traditional musical rhythms, which he called Euclidean rhythms, that originate from all over the world. See Fig. 2 for a depiction of the the maximally even set $J_{16,5}^0$ (see Definition 4.1 and Definition 4.2). For more background on maximally even sets one may consult [4].

Let us note that we take the vertices of C_n as being $V(C_n) = \{0, \dots, n-1\}$ and edges $\{i, (i+1) \bmod n\}$ for each $i \in \{0, \dots, n-1\}$. Fix vertices u and v in C_n . The *clockwise distance* from u to v is the least nonnegative integer congruent to $(v-u) \bmod n$, that is

$$d(u, v) = (v-u) \bmod n.$$

Recall that the *distance* (or *geodesic distance*) between u and v is the length of a shortest path from u to v . For any set $A = \{a_0, \dots, a_{m-1}\}$ of vertices in C_n with $|A| = m \leq n$, arranged in increasing order, we define the A -span of (a_i, a_k) to be $\text{span}_A(a_i, a_k) = (j - i) \bmod m$.

We define two multisets: the k -multispectrum of clockwise distances of A is the multiset

$$\sigma_k^*(A) = [d^*(u, v) : u, v \in A \text{ and } \text{span}_A(u, v) = k]$$

and the k -multispectrum of geodesic distances of A is the multiset

$$\sigma_k(A) = [d(u, v) : u, v \in A \text{ and } \text{span}_A(u, v) = k].$$

For a multiset X we define $\text{supp}(X)$ to be the set whose elements are those of X .

Definition 4.1 (Clough and Douthett [7]). A set A of vertices in C_n with $|A| = m$ is *maximally even* if for each integer k with $1 \leq k \leq m - 1$ the set $\text{supp}(\sigma^*(A))$ consists of either a single integer or two consecutive integers.

Definition 4.2 (Clough and Douthett [7]). Suppose m, n and r are integers with $1 \leq m \leq n$ and $0 \leq r \leq n - 1$. We define

$$J_{n,m}^r = \left\{ \left\lfloor \frac{ni+r}{m} \right\rfloor : i \in \mathbb{Z} \text{ and } 0 \leq i < m \right\}.$$

Sets of the form $J_{n,m}^r$ are called J -representations.

Lemma 4.3 (Clough and Douthett [7, Corollary 1.2]). Suppose k, m, n and r are integers with $1 \leq m \leq n$ and $0 \leq r \leq n - 1$. If $1 \leq k \leq m - 1$ then

$$\text{supp}(\sigma_k^*(J_{n,m}^r)) = \left\{ \left\lfloor \frac{nk}{m} \right\rfloor, \left\lceil \frac{nk}{m} \right\rceil \right\}.$$

Hence $J_{n,m}^r$ is a maximally even subset of C_n .

Indeed, Clough and Douthett [7] showed that a set of vertices A of C_n is maximally even if and only if it is of the form $J_{n,m}^r$ for some appropriate values of m, n and r .

Prior to Clough and Douthett's work on maximal evenness, the following easy lemma was established by Clough and Myerson [8, Lemma 2].

Lemma 4.4. Suppose n is a positive integer, A is a set of vertices in C_n with $|A| = m$ and $1 \leq k \leq m - 1$. Then

$$\sum \sigma_k^*(A) = kn.$$

4.2. A new result on cycles

Before stating our formula for the k -general d -position number of cycles, which is illustrated in Table 2, let us review a proposition from [22].

Remark 4.5. It is stated in [22, Proposition 3.3] that if $n \geq 5$ and $2 \leq d < \lfloor \frac{n}{2} \rfloor$, then

$$\text{gp}_d(C_n) = \begin{cases} 2 \left\lfloor \frac{n}{d+1} \right\rfloor + 1 & \text{if } n \equiv d \pmod{d+1} \\ 2 \left\lfloor \frac{n}{d+1} \right\rfloor & \text{otherwise.} \end{cases} \quad (4.1)$$

Furthermore, if $d \geq \lfloor \frac{n}{2} \rfloor$ then $\text{gp}_d(C_n) = 3$.

Let us point out a minor problem with (4.1). Take $n = 16$ and $d = 5$. Then, according to Proposition (4.1), we have $\text{gp}_5(C_{16}) = 2 \lfloor \frac{16}{6} \rfloor = 4$. However, the set

$$A = J_{16,5}^0 = \left\{ \left\lfloor \frac{16i}{5} \right\rfloor : 0 \leq i < 5 \right\} = \{0, 3, 6, 9, 12\}$$

is a general 5-position set in C_{16} with cardinality 5, thus contradicting (4.1). Let us note that (4.1) can easily be corrected by moving the 2 inside the floor function in the second case (this is a consequence of Theorem 4.6).

Generalizing [22, Proposition 3.3], we obtain the following.

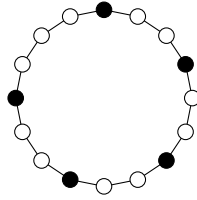


Fig. 2. The maximally even set $J_{16,5}^0$.

Table 2

The quantities $\text{gp}_d^k(C_{14})$ for various values of k and d .

Values of d	Values of k							
	2	3	4	5	6	7	8	9
1	7	14	14	14	14	14	14	14
2	4	9	14	14	14	14	14	14
3	3	7	10	14	14	14	14	14
4	2	5	8	11	14	14	14	14
5	2	4	7	9	11	14	14	14
6	2	4	6	8	10	12	14	14
7	1	3	5	7	8	10	12	14

Theorem 4.6. Suppose d, k and n are positive integers such that $1 \leq d \leq \lfloor \frac{n}{2} \rfloor^1$ and $2 \leq k \leq \lfloor \frac{n}{2} \rfloor + 1$. Then

$$\text{gp}_d^k(C_n) = \begin{cases} n & \text{if } 1 \leq d \leq k - 2. \\ \left\lfloor \frac{(k-1)n}{d+1} \right\rfloor & \text{if } k - 1 \leq d \leq \lfloor \frac{n}{2} \rfloor. \end{cases}$$

Proof. Suppose $1 \leq d \leq k - 2$. It easily follows from Lemma 2.1 that $V(C_n)$ is a k -general d -position set and hence $\text{gp}_d^k(C_n) = n$.

Suppose $k - 2 < d \leq \lfloor \frac{n}{2} \rfloor$. Then, by our assumptions we have $3 \leq k \leq \lfloor \frac{n}{2} \rfloor + 1$ and hence $2 \leq k - 1 \leq \lfloor \frac{n}{2} \rfloor$. Let $m = \lfloor \frac{(k-1)n}{d+1} \rfloor$ and let r be the unique integer such that $(k-1)n = m(d+1) + r$ and $0 \leq r < d+1$.

First we will show that $J_{n,m}^0$ (see Definition 4.2) is a k -general d -position set in C_n . Suppose P is a geodesic of C_n that contains at least k vertices of S . Then, there are $s, s' \in S \cap V(P)$ such that $\text{span}_S(s, s') = k - 1$ and, using Lemma 4.3,

$$d(s, s') = d^*(s, s') \in \text{supp}(\sigma_{k-1}(J_{n,m}^0)) = \left\{ \left\lfloor \frac{(k-1)n}{m} \right\rfloor, \left\lceil \frac{(k-1)n}{m} \right\rceil \right\}.$$

We have

$$d(s, s') \geq \left\lfloor \frac{(k-1)n}{m} \right\rfloor \geq d + 1 + \left\lfloor \frac{r}{m} \right\rfloor \geq d + 1,$$

and hence $\lambda_{C_n}(P) > d$. Thus, $J_{n,m}^0$ is a k -general d -position set in C_n and hence $\text{gp}_d^k(C_n) \geq m$.

To see that $\text{gp}_d^k(C_n) \leq m$, suppose that $S \subseteq V(C_n)$ and $|S| > m$. We will prove that S is not a k -general d -position set in C_n . Let us show that there is some $a \in \sigma_{k-1}^*(S)$ with $a \leq d$. Suppose not, then all elements of $\sigma_{k-1}^*(S)$ are greater than d and since $|S| > m$, by Lemma 4.4, we have

$$(k-1)n = \sum \sigma_{k-1}^*(S) \geq |S|(d+1) \geq (m+1)(d+1).$$

But then $(k-1)n = m(d+1) + r \geq m(d+1) + d + 1$, which implies $r \geq d + 1$, a contradiction. Let $a \in \sigma_{k-1}^*(S)$ be such that $a \leq d$. Then there is a pair $(s, s') \in S \times S$ with $d(s, s') = d^*(s, s') = a \leq d$ such that a shortest path from s to s' contains k elements of S . Hence S is not a k -general d -position set in C_n . $\square$

Corollary 4.7. Suppose d, k, m and n are positive integers such that $1 \leq d \leq \lfloor \frac{n}{2} \rfloor$ and $3 \leq k \leq \lfloor \frac{n}{2} \rfloor + 2$. Given that $\text{gp}_d^k(C_n) = m$, it follows that $J_{n,m}^0$ is a k -general d -position set in C_n .

¹ Let us note that Theorem 4.6 covers all meaningful cases for the values of k and d . For example, if $d > \lfloor \frac{n}{2} \rfloor$, then $d > \text{diam}(C_n)$, and so, by the comments made in the second paragraph after Definition 1.1, $\text{gp}_d^k(C_n) = \text{gp}_{\text{diam}(C_n)}^k(C_n)$.

5. Infinite 2D-grids

Klavžar and Manuel [29] used a corollary of the famous theorem of Erdős and Szekeres to obtain a lower bound on $\text{gp}(P_{\infty}^{\square,2})$. We do the same to obtain a lower bound on $\text{gp}^k(P_{\infty}^{\square,2})$. Recall that $\text{gp}^k(P_{\infty}^{\square,2})$ is the largest cardinality of a set of vertices A in $\text{gp}^k(P_{\infty}^{\square,2})$ with no k vertices in a common geodesic.

Theorem 5.1 (Erdős and Szekeres). *For integers $r, s \geq 2$, any sequence of distinct real numbers with length $(r-1)(s-1)+1$ must contain a monotonically increasing subsequence of length r or a monotonically decreasing subsequence of length s .*

Corollary 5.2. *For every integer $n \geq 2$, every sequence of real numbers $(a_1, \dots, a_N)$ with length $N \geq (n-1)^2 + 1$ contains a monotone subsequence of length n .*

Corollary 5.3. *If $n \geq 1$ is an integer and $S \subseteq \mathbb{R}^2$ with $|S| \geq (n-1)^2 + 1$, then S contains n points that form a monotone sequence.*

We adapt Klavžar and Manuel's notion of monotone-geodesic labeling to that of k -monotone geodesic labeling. Here 3-monotone geodesic labeling is equivalent to monotone-geodesic labeling.

Definition 5.4. Let G be a graph. An injective function $f : V(G) \rightarrow \mathbb{R}^2$ is called a k -monotone-geodesic labeling of G if whenever $v_1, \dots, v_k \in V(G)$ and the sequence $(f(v_1), \dots, f(v_k))$ is monotone, then there is a geodesic $P \in \mathcal{G}(G)$ with $v_1, \dots, v_k \in V(P)$.

Now we connect k -monotone-geodesic labelings with k -general position sets.

Lemma 5.5. *If a graph G admits a k -monotone-geodesic labeling for some integer $k \geq 2$, then $\text{gp}^k(G) \geq (k-1)^2$.*

Proof. Suppose $f : V(G) \rightarrow \mathbb{R}^2$ is a k -monotone-geodesic labeling and $S \subseteq V(G)$ is a k -general position set of G with $|S| \geq (k-1)^2 + 1$. Then, by Corollary 5.3, the set $f(S) \subseteq \mathbb{R}^2$ has a subset T of size k which forms a monotone sequence in $\mathbb{R}^2$. Then $f^{-1}(T)$ has cardinality k and is contained in a single geodesic of G , a contradiction. $\square$

We now obtain the main result of this section. A graph G is called a *grid graph* if it is an induced connected subgraph of $P_{\infty}^{\square,2}$.

Theorem 5.6. *Suppose G is a grid graph and $f : V(P_{\infty}^{\square,2}) \rightarrow \mathbb{R}^2$ is the natural labeling of $P_{\infty}^{\square,2}$. For an integer $k \geq 2$, if G contains $P_{2k-3} \square P_{2k-3}$ as a subgraph and $f \upharpoonright G$ is a k -monotone-geodesic labeling, then $\text{gp}^k(G) = (k-1)^2$.*

Proof. Since $f \upharpoonright G$ is a k -monotone-geodesic labeling, Lemma 5.5 implies that $\text{gp}^k(G) \geq (k-1)^2$. Given a positive integer r , we define

$$A_r = \{(x, y) \in P_{\infty}^{\square,2} : d((0, 0), (x, y)) \in [0, r] \text{ and } d((0, 0), (x, y)) \equiv r \pmod{2}\}.$$

For $(a, b) \in P_{\infty}^{\square,2}$ we let

$$A_r(a, b) = \{(a, b) + (x, y) : (x, y) \in A_r\}.$$

The set $A_{k-2}(\lceil \frac{2k-3}{2} \rceil, \lceil \frac{2k-3}{2} \rceil)$ has cardinality $(k-1)^2$ and is a k -general position set of $P_{2k-3} \square P_{2k-3}$. Since $P_{2k-3} \square P_{2k-3}$ is an isometric subgraph of G , it follows that $A_{k-2}(\lceil \frac{2k-3}{2} \rceil, \lceil \frac{2k-3}{2} \rceil)$ is a k -general position set of G . $\square$

Now we obtain an easy corollary, which was stated in Section 1.

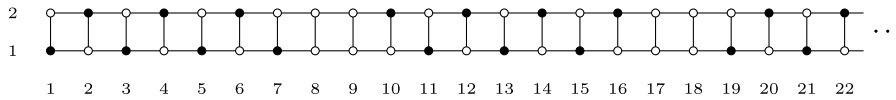
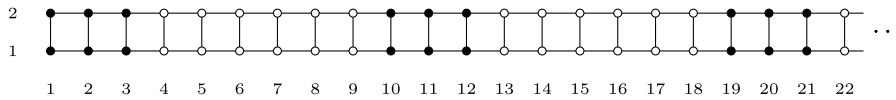
Corollary 5.7. *If $k \geq 2$ is an integer, then*

$$\text{gp}^k(P_{\infty}^{\square,2}) = (k-1)^2.$$

6. Thin finite grids

What would we need to do in order to find $\text{gp}_d^k(P_n \square P_m)$? If a set S of vertices in $P_n \square P_m$ has size $|S| \geq (k-1)^2 + 1$, then, by the results of the previous section, this implies that S has a monotone subset of size k , and then this subset is contained in a geodesic. But, how large must S be to guarantee that S has a monotone subset $M \subseteq S$ of size k such that $\max\{d(u, v) : u, v \in M\} \leq d$? In this section we answer this question when $m = 2$.

For integers d and k with $1 \leq k-1 \leq d$, we will define subsets $A_{d,k}$ and $B_{d,k}$ of $\mathbb{Z}_{\geq 1} \times \{1, 2\}$ that will allow us to easily define k -general d -position subsets of grids of the form $P_n \square P_2$. In what follows, we will see that for every $n \geq 1$, whenever they are defined, the sets $A_{k,d} \cap V(P_n \square P_2)$ and $B_{k,d} \cap V(P_n \square P_2)$ are k -general d -position in $P_n \square P_2$, and furthermore, one of the sets $A_{k,d} \cap V(P_n \square P_2)$ and $B_{k,d} \cap V(P_n \square P_2)$, must have the largest possible cardinality of a k -general d -position set in $P_n \square P_2$.

Fig. 3. The set $A_{5,9} \subseteq \mathbb{Z}_{\geq 1} \times \{1, 2\}$.Fig. 4. The set $B_{5,9} \subseteq \mathbb{Z}_{\geq 1} \times \{1, 2\}$.

Definition 6.1. Suppose d and k are integers with $d \geq 1$ and $k \geq 2$. We define a subset $A_{k,d}$ of $\mathbb{Z}_{\geq 1} \times \{1, 2\}$ as follows. If $d < 2k - 3$ we let $A_{k,d} = \emptyset$. If $d \geq 2k - 3$, for each integer $s \geq 0$, let

$$A_{k,d}(s) = \left\{ \left(i, 1 + \frac{1 + (-1)^{i+s}}{2} \right) : i \in [ds + 1, ds + 2k - 3] \right\}.$$

Define

$$A_{k,d} = \bigcup_{s=0}^{\infty} A_{k,d}(s)$$

(see Fig. 3). For $d \geq k - 2$ we define a subset $B_{k,d}$ of $\mathbb{Z}_{\geq 1} \times \{1, 2\}$ as follows. For each integer s , let

$$B_{k,d}(s) = [ds + 1, ds + k - 2] \times \{1, 2\}.$$

Define

$$B_{k,d} = \bigcup_{s=0}^{\infty} B_{k,d}(s)$$

(see Fig. 4).

First we show that $A_{k,d}$ and $B_{k,d}$ yield k -general d -position sets in grids of the form $P_n \square P_2$, and we take note of the cardinalities of these k -general d -position sets.

Lemma 6.2. Suppose k and d are integers with $d \geq 1$ and $k \geq 2$.

- (1) Suppose $d \geq 2k - 3$. Then $A_{k,d}$ is a k -general d -position set in $P_{\infty} \square P_2$. Furthermore, for every integer $n \geq 1$, $A_{k,d} \cap V(P_n \square P_2)$ is a k -general d -position set in $P_n \square P_2$ with

$$|A_{k,d} \cap V(P_n \square P_2)| = (2k - 3) \left\lfloor \frac{n}{d} \right\rfloor + \min(n \bmod d, 2k - 3).$$

- (2) Suppose $d \geq k - 2$. Then $B_{k,d}$ is a k -general d -position set in $P_{\infty} \square P_2$. Furthermore, for every integer $n \geq 1$, $B_{k,d} \cap V(P_n \square P_2)$ is a k -general d -position set in $P_n \square P_2$ with

$$|B_{k,d} \cap V(P_n \square P_2)| = 2(k - 2) \left\lfloor \frac{n}{d} \right\rfloor + 2 \min(n \bmod d, k - 2).$$

Proof. Let us prove that, when they are defined $A_{k,d}$ and $B_{k,d}$ are k -general d -position sets in $P_{\infty} \square P_2$. The rest follows easily from Lemma 2.1 as well as the definitions of $A_{k,d}$ and $B_{k,d}$. When $A_{k,d}$ and $B_{k,d}$ are defined, for each $s \geq 0$ we have $|A_{k,d}(s)| = 2k - 3$ and $|B_{k,d}(s)| = 2(k - 2) = 2k - 4$. For any geodesic P of $P_{\infty} \square P_2$, it follows that for every $s \geq 0$ we have $|A_{k,d}(s) \cap V(P)| \leq \left\lceil \frac{2k-3}{2} \right\rceil = k - 1$ and $|B_{k,d}(s) \cap V(P)| \leq k - 1$. Suppose s and s' are integers such that $s, s' \geq 0$ and $|s - s'| \leq 1$. Suppose P is a geodesic of $P_{\infty} \square P_2$ with length $\lambda(P) = d$. If $|A_{k,d}(s) \cap V(P)| = a \leq k - 1$ then $|A_{k,d}(s') \cap V(P)| = k - 1 - a$. Thus, it follows that $|A_{k,d} \cap V(P)| \leq k - 1$. Similarly $|B_{k,d} \cap V(P)| \leq k - 1$. Therefore $A_{k,d}$ and $B_{k,d}$ are k -general d -position sets in $P_{\infty} \square P_2$. $\square$

Let us state the main result of the current section, which easily follows from Theorems 6.4, 6.5 and 6.8.

Theorem 6.3. Suppose d, k, n are positive integers with $k \geq 2$ and $d \geq k - 1$. Then

$$\text{gp}_d^k(P_n \square P_2) = \max(|A_{k,d} \cap V(P_n \square P_2)|, |B_{k,d} \cap V(P_n \square P_2)|).$$

The next theorem easily implies Theorem 6.3 in the case where $k = 2$.

Theorem 6.4. Suppose d and n are integers such that $d \geq 1$ and $n \geq 1$. Then

$$\text{gp}_d^2(P_n \square P_2) = \left\lceil \frac{n}{d} \right\rceil.$$

Proof. Since $d \geq 2(2) - 3 = 1$ the set $A_{2,d}$ is nonempty, and by Lemma 6.2, $A_{2,d} \cap V(P_n \square P_2)$ is in 2-general d -position in $P_n \square P_2$. Also by Lemma 6.2,

$$|A_{2,d} \cap V(P_n \square P_2)| = \left\lfloor \frac{n}{d} \right\rfloor + \min(n \bmod d, 1) \geq \left\lceil \frac{n}{d} \right\rceil.$$

Thus $\text{gp}_d^2(P_n \square P_2) \geq \left\lceil \frac{n}{d} \right\rceil$.

Suppose A is a 2-general d -position set in $P_n \square P_2$ with $|A| > \left\lceil \frac{n}{d} \right\rceil$. Let $q = \left\lfloor \frac{n}{d} \right\rfloor$ and $r = n \bmod d$. Then $n = qd + r$ where $0 \leq r < d$. For each $i \in \{0, \dots, q-1\}$ let $X_i = [di+1, d(i+1)] \times \{1, 2\}$, and if $r > 0$ let $X_q = [dq+1, dq+r] \times \{1, 2\}$. Since $|A| > \left\lceil \frac{n}{d} \right\rceil$ at least one part of the partition defined by the X_i 's must contain at least two vertices of A . But the diameter of each induced isometric subgraph $(P_n \square P_2)[X_i]$ is at most d , which implies that A is not in 2-general d -position in $P_n \square P_2$. $\square$

Next we obtain the desired result in the case where $k = 3$.

Theorem 6.5. Suppose d and n are integers such that $d \geq 1$ and $n \geq 1$.

(1) If $n \leq d$ then

$$\text{gp}_d^3(P_n \square P_2) = \begin{cases} 2 & \text{if } n = 1 \\ \min(n, 3) & \text{if } n \geq 2. \end{cases}$$

(2) If $n \geq d+1$ then

$$\text{gp}_d^3(P_n \square P_2) = \begin{cases} 2 \left\lfloor \frac{n}{2} \right\rfloor + 2 \min(n \bmod 2, 1) & \text{if } d = 2 \\ 3 \left\lfloor \frac{n}{d} \right\rfloor + \min(n \bmod d, 3) & \text{if } d \geq 3. \end{cases}$$

Proof. Let us prove (1). Suppose $1 = n \leq d$. In $P_1 \square P_2$, there are no geodesics with length greater than $d \geq 1$, so $\text{gp}_d^3(P_1 \square P_2) = \text{gp}^3(P_1 \square P_2) = \text{gp}^3(P_2) = 2$. Suppose $n = 2$. Similarly, in $P_2 \square P_2$, there are no geodesics with length greater than $d \geq 2$ and hence $\text{gp}_d^3(P_2 \square P_2) = \text{gp}^3(P_2 \square P_2) = 2$.

Suppose $n \leq d$ and $n \geq 3$. Then, by Lemma 6.2, $A_{3,d} \cap V(P_n \square P_2)$ is a 3-general d -position set in $P_n \square P_2$ with

$$|A_{3,d} \cap V(P_n \square P_2)| = 3 \left\lfloor \frac{n}{d} \right\rfloor + \min(n \bmod d, 3) = 3.$$

Thus, $\text{gp}_d^3(P_n \square P_2) \geq 3$. Let us prove that $\text{gp}_d^3(P_n \square P_2) \leq 3$. For the sake of contradiction suppose A is a k -general d -position set in $P_n \square P_2$ with $|A| > 3$. Without loss of generality we can assume that $|A| = 4$. Since $[1, n] \times \{1\}$ and $[1, n] \times \{2\}$ are vertex sets of geodesics of length less than d , it follows that $|A \cap ([1, n] \times \{1\})| = 2$ and $|A \cap ([1, n] \times \{2\})| = 2$. Let i_0 be the least $i \in [1, n]$ such that $A \cap \{(i, 1), (i, 2)\} \neq \emptyset$. Without loss of generality, suppose $(i_0, 1) \in A$. Then $\{(i_0, 1)\} \cup ([i_0, n] \times \{2\})$ is the vertex set of a geodesic of $P_n \square P_2$ with length at most d containing 3 vertices from A , a contradiction. This establishes (1).

Now let us prove (2). For the remainder of the proof, let $q = \left\lfloor \frac{n}{d} \right\rfloor$ and $r = n \bmod d$. Suppose $n \geq d+1$ and $d = 2$. Notice that $r \in \{0, 1\}$. By Lemma 6.2, $B_{3,2} \cap V(P_n \square P_2)$ is a 3-general 2-position set in $P_n \square P_2$ with

$$|B_{3,2} \cap V(P_n \square P_2)| = 2 \left\lfloor \frac{n}{2} \right\rfloor + 2 \min(r, 1).$$

Thus $\text{gp}_2^3(P_n \square P_2) \geq 2 \left\lfloor \frac{n}{2} \right\rfloor + 2 \min(r, 1)$. To show that $\text{gp}_2^3(P_n \square P_2) \leq 2 \left\lfloor \frac{n}{2} \right\rfloor + 2 \min(r, 1)$ we will use Lemma 2.7. For $i \in \{0, \dots, q-1\}$, let $X_i = [2i+1, 2i+2] \times \{1, 2\}$. If $r = 1$, let $X_q = \{2q+1\} \times \{1, 2\}$ and if $r = 0$ let $X_q = \emptyset$. For $i \in \{0, \dots, q\}$, let $H_i = (P_n \square P_2)[X_i]$ and note that H_i is an isometric subgraph of $P_n \square P_2$. Furthermore, $V(P_n \square P_2) = \bigcup_{i=0}^q V(H_i)$. Hence, by Lemma 2.7, and by using the fact that $H_i \cong P_2 \square P_2$ for $i \in \{0, \dots, q-1\}$ and $H_q \cong P_r \square P_2$, we obtain

$$\text{gp}_2^3(P_n \square P_2) \leq \sum_{i=0}^q \text{gp}_2^3(H_i) = \text{gp}_2^3(P_2 \square P_2)q + \text{gp}_2^3(P_r \square P_2) = 2q + 2m = 2 \left\lfloor \frac{n}{2} \right\rfloor.$$

Suppose $n \geq d+1$ and $d \geq 3$. By Lemma 6.2, $A_{3,d}$ is a 3-general d -position set in $P_n \square P_2$ with

$$|A_{3,d} \cap V(P_n \square P_2)| = 3 \left\lfloor \frac{n}{d} \right\rfloor + \min(r, 3).$$

Thus $\text{gp}_d^3(P_n \square P_2) \leq 3 \left\lfloor \frac{n}{d} \right\rfloor + \min(r, 3)$. Now we will prove the reverse direction of this inequality.

For the remainder of the proof, let us fix some notation. For $i \in \{0, \dots, q-1\}$, let $X_i = [di+1, di+d] \times \{1, 2\}$. If $r > 0$, let $X_q = [dq+1, dq+r] \times \{1, 2\}$ and if $r = 0$, let $X_q = \emptyset$. For $i \in \{0, \dots, q-1\}$ let $H_i = (P_n \square P_2)[X_i]$.

First, suppose $r \neq 1$. By Lemma 2.7, it follows from (1) that

$$\begin{aligned} \text{gp}_d^3(P_n \square P_2) &\leq \sum_{i=0}^q \text{gp}_d^3(H_i) = \text{gp}_d^3(P_d \square P_2)q + \text{gp}_d^3(P_r \square P_2) \\ &= \min(d, 3)q + \min(r, 3) \\ &= 3q + m, \end{aligned}$$

as desired.

Suppose $r = 1$. Then $V(H_q) = X_q = \{dq+1\} \times \{1, 2\}$ contains two vertices. We must show that $\text{gp}_d^3(P_n \square P_2) \leq 3q+1$. For the sake of contradiction, suppose A is a 3-general d -position set in $P_n \square P_2$ with $|A| > 3q+1$. Without loss of generality suppose $|A| = 3q+2$. For $i \in \{0, \dots, q-1\}$ we have $\text{gp}_d^3(H_i) = \text{gp}_d^3(P_d \square P_2) = 3$ and thus, by Lemma 2.6, $|A \cap X_i| = 3$. This implies that $A \cap X_q = 2$. By the pigeonhole principle, we can let $j \in \{1, 2\}$ be such that $[d(q-1)+1, dq] \times \{j\}$ contains two vertices of A . But then $[d(q-1)+1, dq+1] \times \{j\}$ is the vertex set of a geodesic of length d that contains three vertices of A , a contradiction. $\square$

Before we obtain the value of $\text{gp}_d^k(P_n \square P_2)$ for $k \geq 4$, we need a few lemmas.

Lemma 6.6. Suppose d and k are positive integers with $k \geq 4$ and $d \geq 2k-3$. Let A be any k -general d -position set in $P_d \square P_2$ with $|A| = 2k-3$. For all $i \in [1, d]$ we have that

(1) if $j \in \{1, 2\}$ is such that $|A \cap ([1, d] \times \{j\})| = k-1$, then

$$|A \cap ([1, i] \times \{j\})| \leq \left\lceil \frac{i}{2} \right\rceil$$

and

(2) if $j \in \{1, 2\}$ is such that $|A \cap ([1, d] \times \{j\})| = k-2$, then

$$|A \cap ([1, i] \times \{j\})| \leq \left\lfloor \frac{i}{2} \right\rfloor.$$

Proof. Without loss of generality, suppose $|A \cap ([1, d] \times \{1\})| = k-1$ and $|A \cap ([1, d] \times \{2\})| = k-2$. We prove that (1) and (2) hold for all desired values of i by simultaneous induction.

Notice that $(1, 2) \notin A$ because if $(1, 2) \in A$ then $\{(1, 2)\} \cup ([1, d] \times \{1\})$ would be the vertex set of a geodesic of $P_d \square P_2$ of length d containing k vertices of A , a contradiction. This implies that $|A \cap ([1, 1] \times \{2\})| \leq \lfloor \frac{1}{2} \rfloor = 0$ and $|A \cap ([1, 2] \times \{2\})| \leq \lfloor \frac{2}{2} \rfloor = 1$.

We must have $|A \cap ([1, 2] \times \{1\})| \leq 1$ because otherwise, if $|A \cap ([1, 2] \times \{1\})| = 2$ then, since $(1, 2) \notin A$ by the previous case, it follows that

$$([1, 2] \times \{1\}) \cup ([2, d] \times \{2\})$$

is the vertex set of a geodesic of $P_d \square P_2$ containing k vertices of A , a contradiction. This implies that $|A \cap ([1, 1] \times \{1\})| \leq \lceil \frac{1}{2} \rceil = 1$ and $|A \cap ([1, 2] \times \{1\})| \leq \lceil \frac{2}{2} \rceil = 1$.

Suppose (1) and (2) hold for i . Since $|A \cap ([1, i] \times \{1\})| \leq \lceil \frac{i}{2} \rceil$, we must have $|A \cap ([1, i+1] \times \{2\})| \leq \lfloor \frac{i+1}{2} \rfloor$ because otherwise, if $|A \cap ([1, i+1] \times \{2\})| > \lfloor \frac{i+1}{2} \rfloor$, then there is a geodesic P of $P_d \square P_2$ with

$$V(P) = ([1, i+1] \times \{2\}) \cup ([i+1, d] \times \{1\})$$

such that $V(P)$ contains at least

$$k-1 - \left\lceil \frac{i}{2} \right\rceil + \left\lfloor \frac{i+1}{2} \right\rfloor + 1 = k$$

vertices of A , a contradiction.

Similarly, since $|A \cap ([1, i] \times \{2\})| \leq \lfloor \frac{i}{2} \rfloor$, we must have $|A \cap ([1, i+1] \times \{1\})| \leq \lceil \frac{i+1}{2} \rceil$ because otherwise, if $|A \cap ([1, i+1] \times \{1\})| \geq \lceil \frac{i+1}{2} \rceil$ then there is a geodesic P of $P_d \square P_2$ with

$$V(P) = ([1, i+1] \times \{1\}) \cup ([i+1, d] \times \{2\})$$

such that $V(P)$ contains at least

$$k-2 - \left\lfloor \frac{i}{2} \right\rfloor + \left\lceil \frac{i+1}{2} \right\rceil + 1 = k$$

vertices of A , a contradiction. $\square$

Lemma 6.7. Suppose d, k and r are positive integers such that $k \geq 4$, $r \geq 1$ and $d \geq 2k - 3$. If A is any set of vertices in $P_{d+r} \square P_2$ with $|A \cap ([1, d] \times \{1, 2\})| \geq 2k - 3$ and $|A \cap ([d + 1, d + r] \times \{1, 2\})| \geq r + 1$, then A is not a k -general d -position set in $P_{d+r} \square P_2$.

Proof. For the sake of contradiction, suppose A is a k -general d -position set in $P_{d+r} \square P_2$ satisfying the stated hypotheses.

Without loss of generality suppose $|A \cap ([1, d] \times \{1\})| \geq k - 1$ and $|A \cap ([1, d] \times \{2\})| \geq k - 2$. It follows by the pigeonhole principle that either

$$|A \cap ([d + 1, d + r] \times \{1\})| \geq \left\lceil \frac{r + 1}{2} \right\rceil$$

or

$$|A \cap ([d + 1, d + r] \times \{2\})| \geq \left\lceil \frac{r + 1}{2} \right\rceil.$$

Suppose $|A \cap ([d + 1, d + r] \times \{1\})| \geq \left\lceil \frac{r + 1}{2} \right\rceil$. Then there is a geodesic P of $P_{d+r} \square P_2$ with

$$V(P) = [r, d + r] \times \{1\}$$

which has length d . By Lemma 6.6, the set $[1, r - 1] \times \{1\}$ contains at most $\left\lceil \frac{r - 1}{2} \right\rceil$ vertices of A . Therefore, P contains at least

$$k - 1 - \left\lceil \frac{r - 1}{2} \right\rceil + \left\lceil \frac{r + 1}{2} \right\rceil = k$$

vertices of A , a contradiction.

Now suppose $|A \cap ([d + 1, d + r] \times \{2\})| \geq \left\lceil \frac{r + 1}{2} \right\rceil$. We can assume, without loss of generality, that r is even because when r is odd we have $\left\lceil \frac{r + 1}{2} \right\rceil = \left\lfloor \frac{r + 1}{2} \right\rfloor$, and thus the assumption $|A \cap ([d + 1, d + r] \times \{1\})| \geq \left\lceil \frac{r + 1}{2} \right\rceil$ of the previous case holds. By Lemma 6.6, the set $[1, r - 1] \times \{2\}$ contains at most $\left\lfloor \frac{r - 1}{2} \right\rfloor$ vertices of A . Thus, there is a geodesic P' of $P_{d+r} \square P_2$ with

$$V(P') = [r, d + r] \times \{2\}$$

such that $V(P')$ contains at least

$$k - 2 + \left\lfloor \frac{r - 1}{2} \right\rfloor + \left\lceil \frac{r + 1}{2} \right\rceil = k$$

vertices of A , where the last equality holds because r is even. Since P' has length d we obtain a contradiction. $\square$

Now we obtain the values of $\text{gp}_d^k(P_n \square P_2)$ for $k \geq 4$. By using the formulas for the cardinalities of $A_{k,d} \cap V(P_n \square P_2)$ and $A_{k,d} \cap V(P_n \square P_2)$, and by considering cases, it is easy to see that Theorem 6.3 is equivalent to the next theorem for $k \geq 4$; we restate Theorem 6.3 in this way because this will facilitate our proof.

Theorem 6.8. Suppose d, k and n are positive integers such that $k \geq 4$ and $d \geq k - 1$.

(1) If $n \leq d$ then

$$\text{gp}_d^k(P_n \square P_2) = \begin{cases} 2 \min(n, k - 2) & \text{if } 1 \leq n \leq 2k - 4 \\ 2k - 3 & \text{if } n \geq 2k - 3. \end{cases}$$

(2) If $n \geq d + 1$ and $d < 2k - 3$ then

$$\text{gp}_d^k(P_n \square P_2) = (2k - 4) \left\lfloor \frac{n}{d} \right\rfloor + 2 \min(n \bmod d, k - 2).$$

(3) Suppose $n \geq d + 1$ and $d \geq 2k - 3$.

(a) Suppose $n \bmod d \leq k - 2$.

(i) If $n \bmod d \leq \left\lfloor \frac{n}{d} \right\rfloor$ then

$$\text{gp}_d^k(P_n \square P_2) = (2k - 3) \left\lfloor \frac{n}{d} \right\rfloor + n \bmod d.$$

(ii) If $n \bmod d > \left\lfloor \frac{n}{d} \right\rfloor$ then

$$\text{gp}_d^k(P_n \square P_2) = (2k - 4) \left\lfloor \frac{n}{d} \right\rfloor + 2(n \bmod d).$$

(b) Suppose $k - 2 < n \bmod d < 2k - 3$.

(i) If $2k - 4 - n \bmod d \leq \lfloor \frac{n}{d} \rfloor$ then

$$\text{gp}_d^k(P_n \square P_2) = (2k - 3) \left\lfloor \frac{n}{d} \right\rfloor + n \bmod d.$$

(ii) If $2k - 4 - n \bmod d > \lfloor \frac{n}{d} \rfloor$ then

$$\text{gp}_d^k(P_n \square P_2) = (2k - 4) \left(\left\lfloor \frac{n}{d} \right\rfloor + 1 \right).$$

(c) If $n \bmod d \geq 2k - 3$ then

$$\text{gp}_d^k(P_n \square P_2) = (2k - 3) \left(\left\lfloor \frac{n}{d} \right\rfloor + 1 \right).$$

Proof. For (1), suppose $n \leq d$. To prove the formula in (1), let us consider a few cases. The largest cardinality of a vertex set of a geodesic in $P_n \square P_2$ is $n + 1$, so if $1 \leq n \leq k - 2$, there are no geodesics of $P_n \square P_2$ that contain k vertices of $V(P_n \square P_2)$, which implies that $A_n = V(P_n \square P_2)$ is a k -general d -position set in $P_n \square P_2$ with $|A_n| = 2n$. Therefore, when $1 \leq n \leq k - 2$ we have $\text{gp}_d^k(P_n \square P_2) = 2n$.

Suppose $k - 1 \leq n \leq 2k - 4$. By Lemma 6.2, $B_{k,d} \cap V(P_n \square P_2)$ is a k -general d -position set in $P_n \square P_2$ with cardinality

$$|B_{k,d} \cap V(P_n \square P_2)| = 2(k - 2) \left\lfloor \frac{n}{d} \right\rfloor + 2 \min(n \bmod d, k - 2) = 2(k - 2)$$

(where the last equality can be seen to hold by considering cases in which $n < d$ and $n = d$). Thus we have $\text{gp}^k(P_n \square P_2) \geq 2k - 4$. To show that $\text{gp}^k(P_n \square P_2) \leq 2k - 4$, assume for the sake of contradiction that A is a k -general d -position set of vertices in $P_n \square P_2$ with $|A| > 2k - 4$. Since $n \leq 2(k - 2)$, it follows by the pigeonhole principle that we may fix an $i \in [1, n]$ such that $|A \cap (\{i\} \times \{1, 2\})| = 2$. There are exactly two geodesics, P_1 and P_2 of $P_n \square P_2$ whose vertex sets contain $\{i\} \times \{1, 2\}$ and that have maximal length $\lambda(P_1) = \lambda(P_2) = n$. Since $|A \setminus (\{i\} \times \{1, 2\})| > 2k - 6$, and since all of the vertices in $A \setminus (\{i\} \times \{1, 2\})$ must live in P_1 or P_2 , it follows by the pigeonhole principle that for some $\ell \in \{1, 2\}$ we have

$$|(A \setminus (\{i\} \times \{1, 2\})) \cap V(P_\ell)| \geq k - 2$$

and hence

$$|A \cap V(P_\ell)| \geq k,$$

where $\lambda(P_\ell) = n \leq d$, a contradiction.

To finish (1), suppose $2k - 3 \leq n \leq d$. We must show that $\text{gp}_d^k(P_n \square P_2) = 2k - 3$. By Lemma 6.2, $A_{k,d} \cap V(P_n \square P_2)$ is a k -general d -position set in $P_n \square P_2$ with cardinality

$$|A_{k,d} \cap V(P_n \square P_2)| = (2k - 3) \left\lfloor \frac{n}{d} \right\rfloor + \min(n \bmod d, 2k - 3) = 2k - 3$$

(where the last inequality can be seen to hold by considering cases in which $n < d$ and $n = d$). Thus, it follows that $\text{gp}^k(P_n \square P_2) \geq 2k - 3$. To show that $\text{gp}^k(P_n \square P_2) \leq 2k - 3$, assume for the sake of contradiction that A is a k -general d -position set of vertices in $P_n \square P_2$ with $|A| > 2k - 3$. Without loss of generality we can assume that $|A| = 2(k - 1)$. If $|A \cap ([1, n] \times \{j\})| \geq k$ for some $j \in \{1, 2\}$, then $[1, n] \times \{j\}$ is the vertex set of a geodesic containing at least k elements of A that has length $n - 1 < d$, a contradiction. Thus, we can assume that

$$|A \cap ([1, n] \times \{j\})| = k - 1$$

for $j \in \{1, 2\}$. Let i_0 be the least $i \in [1, n]$ such that $A \cap (\{i\} \times \{1, 2\}) \neq \emptyset$. Without loss of generality, suppose $(i_0, 1) \in A$. Then it follows that there is a geodesic P with

$$V(P) = \{(i_0, 1), (i_0, 2)\} \cup \{(i, 2) : i_0 \leq i \leq n\},$$

which contains k members of A and has length at most $n \leq d$, a contradiction.

For (2), suppose $n \geq d + 1$ and $d < 2k - 3$. By Lemma 6.2, $B_{k,d} \cap V(P_n \square P_2)$ is a k -general d -position set in $P_n \square P_2$ with cardinality

$$|B_{k,d} \cap V(P_n \square P_2)| = 2(k - 2) \left\lfloor \frac{n}{2} \right\rfloor + 2 \min(n \bmod d, k - 2)$$

and thus

$$\text{gp}_d^k(P_n \square P_2) \geq 2(k - 2) \left\lfloor \frac{n}{2} \right\rfloor + 2 \min(n \bmod d, k - 2).$$

Let us show that

$$\text{gp}_d^k(P_n \square P_2) \leq 2(k - 2) \left\lfloor \frac{n}{2} \right\rfloor + 2 \min(n \bmod d, k - 2).$$

Let $q = \lfloor \frac{n}{d} \rfloor$ and let $r = n \bmod d$ be the unique integer such that $n = dq + r$ where $0 \leq r < d$. For each $i \in \{0, \dots, q-1\}$ let $X_i = [di+1, di+d] \times \{1, 2\}$. If $r = 0$ let $X_i = \emptyset$ and if $r > 0$ let $X_i = [dq+1, n] \times \{1, 2\}$. For $i \in \{0, \dots, q\}$ let $H_i = (P_n \square P_2)[X_i]$ be the subgraph induced by X_i and note that H_i is an isometric subgraph of $P_n \square P_2$. If $0 \leq i \leq q-1$ then $H_i \cong P_d \square P_2$ and $H_q \cong P_r$ (where P_0 is the empty graph). By (1) (which we already proved above), it follows from $k-1 \leq d < 2k-3$ that $\text{gp}_d^k(P_d \square P_2) = 2(k-2)$. Also by (1), since $0 \leq r < d < 2k-3$, we see that $\text{gp}_d^k(P_r \square P_2) = 2 \min(r, k-2)$. Therefore, by Lemma 2.7 we obtain

$$\begin{aligned} \text{gp}_d^k(P_n \square P_2) &\leq \sum_{i=0}^q \text{gp}_d^k(H_i) \\ &= \text{gp}_d^k(P_d \square P_2)q + \text{gp}_d^k(P_r \square P_2) \\ &= 2(k-2) \left\lfloor \frac{n}{d} \right\rfloor + 2 \min(n \bmod d, k-2). \end{aligned}$$

Let us prove (3). Suppose $n \geq d+1$ and $d \geq 2k-3$. By Lemma 6.2, both $A_{k,d} \cap V(P_n \square P_s)$ and $B_{k,d} \cap V(P_n \square P_2)$ are k -general d -position sets and thus, given the cardinalities stated in Lemma 6.2, we obtain

$$\begin{aligned} \text{gp}_d^k(P_n \square P_2) &\geq \max(2(k-2) \left\lfloor \frac{n}{d} \right\rfloor + 2 \min(n \bmod d, k-2), \\ &\quad (2k-3) \left\lfloor \frac{n}{d} \right\rfloor + \min(n \bmod d, 2k-3)). \end{aligned}$$

This implies that, in each sub-case of (3), $\text{gp}_d^k(P_n \square P_r)$ is greater than or equal to the right-hand-side of the stated equation.

Let us prove the reverse inequality in each case of (3). For notational simplicity, let $q = \lfloor \frac{n}{d} \rfloor$ and $r = n \bmod d$.

Let us now finish the proof of (3ai) and (3bi) with a single argument. Suppose $r < 2k-3$. We must prove that

$$\text{gp}_d^k(P_n \square P_2) \leq (2k-3)q + r.$$

For the sake of contradiction, suppose A is a k -general d -position set in $P_n \square P_2$ such that $|A| > (2k-3)q + r$. Without loss of generality we can assume that $|A| = (2k-3)q + r + 1$. By Lemma 2.6 and (1), it follows that $|A \cap X_i| \leq 2k-3$ for $i \in \{0, \dots, q-1\}$ and thus $|A \cap X_q| = r+1$. Lemma 2.6 implies that $A \cap ([dq-d+1, dq+r] \times \{1, 2\})$ is a k -general d -position set in the induced subgraph

$$(P_n \square P_2)[[dq-d+1, dq+r] \times \{1, 2\}].$$

But this contradicts Lemma 6.7.

Let us finish the proof of (3aii). Suppose $r \leq k-2$ and $r > q$. For the sake of contradiction, suppose A is a k -general d -position set in $P_n \square P_2$ with $|A| > (2k-4)q + 2r$. Without loss of generality, suppose that $|A| = (2k-4)q + 2r + 1$. It follows from (1) and the fact that for $i \in \{0, \dots, q-1\}$, $A \cap X_i$ is a k -general d -position set in H_i , that

$$|A \cap ([dq+1, dq+r] \times \{1, 2\})| \geq |A| - (2k-3)q = 2r + 1 - q \geq r + 2.$$

The rest of the argument is the same as that of (3ai) and (3bi).

Let us finish the proof of (3bii). Suppose $2k-4-r > q$. For the sake of contradiction, suppose A is a k -general d -position set in $P_n \square P_2$ with $|A| = (2k-4)(q+1) + 1 = (2k-4)q + 2k-3$. It follows from (1) and the fact that for $i \in \{0, \dots, q-1\}$, $A \cap X_i$ is a k -general d -position set in H_i , that $|A \cap X_i| \leq 2k-3$ for $i \in \{0, \dots, q-1\}$ and thus

$$|A \cap ([dq+1, dq+r] \times \{1, 2\})| \geq |A| - (2k-3)q = -q + 2k-3 \geq r+1.$$

The rest of the argument is the same as that of (3ai), (3bi) and (3aii).

Finally, what remains to prove of (3c) is a straightforward consequence of Lemma 2.7. $\square$

Remark 6.9. Let us make note of a curious identity that hints at some connection between the two-dimensional k -general d -position subsets of $P_n \square P_2$ on the one hand and the one-dimensional $2k-2$ -general $d-1$ -position, as well as $k-1$ -general $d-1$ -position subsets of P_n on the other. By inspecting the formulas above, it is apparent that when $n \geq d+1$ and $d \geq 2k-3$ we have

$$\text{gp}_d^k(P_n \square P_2) = \max(\text{gp}_{d-1}^{2k-2}(P_n), 2\text{gp}_{d-1}^{k-1}(P_n)).$$

Is there a way to use the formula for $\text{gp}_d^k(P_n)$ (Theorem 3.3) to prove the formulas for $\text{gp}_d^k(P_n \square P_2)$ in Theorem 6.3 and Theorem 6.8? If so, this might provide a strategy for proving formulas for $\text{gp}_d^k(P_n \square P_3)$.

7. Open problems

Many questions regarding k -general d -position subsets of graphs remain open. Let us provide some background information in order to motivate our first list of open problems. In Section 6, we found a formula for $\text{gp}_d^k(P_n \square P_2)$. The general position number of toroidal graphs $\text{gp}(C_n \square C_m)$ has been studied in [24,26]. Klavzar et. al. [24] provided information on the general position number $\text{gp}(P_n \square C_m)$ of cylinders. The general position number of the n -dimensional infinite grid $\mathcal{P}_\infty^{\square, n} = P_\infty \square \dots \square P_\infty$ has been studied in [25].

Open Problems 7.1. Find formulas for the k -general d -position numbers of the following graphs in terms of the relevant parameters.

- (1) $P_n \square P_m$ for $m > 2$ (2-dimensional finite grids, see Remark 6.9)
- (2) $P_{n_1} \square \cdots \square P_{n_\ell}$ (ℓ -dimensional finite grids)
- (3) $C_n \square C_m$ (toroidal graphs)
- (4) $C_{n_1} \square \cdots \square C_{n_\ell}$ (ℓ -dimensional toroidal graphs)
- (5) $P_n \square C_m$ (cylinders)
- (6) $P_\infty^{\square, n}$ (n -dimensional infinite grid)

Considering Corollary 5.7 and the results of [25], we make the following conjecture.

Conjecture 7.2. $\text{gp}^k(P_\infty^{\square, n}) = (k - 1)^{2^{n-1}}$.

Klavžar, Rall and Yero [22] proved the following structural characterization of general d -position sets. Recall that subgraphs H_1 and H_2 of a graph G are *parallel*, denoted by $H_1 \parallel H_2$, if for every pair of vertices $u_1 \in V(H_1)$ and $u_2 \in V(H_2)$ we have $d_G(u_1, u_2) = d_G(H_1, H_2)$.

Theorem 7.3 ([22]). *Let G be a connected graph and let $d \geq 2$ be an integer. Then $S \subseteq V(G)$ is a general d -position set if and only if the following conditions hold:*

- (1) *The induced subgraph $G[S]$ is a disjoint union of complete graphs $Q_1, \dots, Q_\ell$.*
- (2) *If $Q_i \not\parallel Q_j$ where $i \neq j$, then $d_G(Q_i, Q_j) \geq d$.*
- (3) *If $d_G(Q_i, Q_j) + d_G(Q_j, Q_k) = d_G(Q_i, Q_k)$ for $\{i, j, k\} \subseteq \{1, \dots, \ell\}$ with $|\{i, j, k\}| = 3$, then $d_G(Q_i, Q_k) > d$.*

Question 7.4. *Can one characterize k -general d -position subsets of connected graphs in a way which is similar to the characterization of general d -position sets given in Theorem 7.3?*

Klavžar, Rall and Yero [22, Section 2] showed that one can construct graphs with various configurations of strict and non-strict inequalities in the chain $\text{gp}_2(G) \geq \text{gp}_3(G) \geq \cdots \geq \text{gp}_{\text{diam}(G)}(G)$. For example, if T_n is the tree obtained from a path P_{n+1} on $n + 1$ vertices by attaching two leaves to each of its internal vertices then

$$\text{gp}_2(T_n) = \text{gp}_3(T_n) = \cdots = \text{gp}_n(T_n).$$

Furthermore, they construct a graph H_t with diameter t such that

$$\text{gp}_2(H_t) > \text{gp}_3(H_t) > \cdots > \text{gp}_t(H_t).$$

Question 7.5. *To what extent can one construct graphs for which various configurations of strict and non-strict inequalities among the values of $\text{gp}_d^k(G)$ are realized? Recall that the lattice of inequalities in Fig. 1 holds in general. For example, is there a graph for which all the inequalities in Fig. 1 are strict?*

Klavžar, Rall and Yero showed that the general d -position problem is NP-complete [22, Theorem 2.2]. We end with a conjecture regarding the computational complexity of the following decision problem related to finding the k -general d -position number of graphs.

k -GENERAL d -POSITION PROBLEM

Input : A graph G , an integer $k \geq 2$, an integer $d \geq 1$
and a positive integer r .

Question : Is $\text{gp}_d^k(G)$ greater than r ?

Conjecture 7.6. *For any integers k and d with $k \geq 2$ and $d \geq k - 1$, the k -GENERAL d -POSITION PROBLEM is NP-complete.*

We close the paper with another family of open problems concerning generalizations of the notions of mutual visibility for graphs, as introduced in [10]. For example, in [5] the following notions are defined. For a given integer $d \geq 1$ and a set of vertices S of a graph G , two vertices $u, v \in V(G)$ are called S_d -visible if there is a shortest path P from u to v in G of length at most d such that $S \cap (V(P) \setminus \{u, v\}) = \emptyset$. The set S is a d -distance mutual-visibility set if every two vertices of S are S_d -visible. The cardinality of a largest d -distance mutual-visibility set in G is the d -distance mutual-visibility number of G , denoted by $\mu_k(G)$.

Just as we generalized the notion of general d -position set of vertices in the current article, one could generalize d -distance mutual-visibility sets by introducing another parameter. For integers $d \geq 1$ and $k \geq 2$, and a set of vertices S of a graph G , we say that two vertices $u, v \in V(G)$ are called S_d^k -visible if there is a shortest path P from u to v in G of length

at most d such that $|S \cap (V(P) \setminus \{u, v\})| \leq k$. The set S is a d -distance k -mutual-visibility set if every two vertices of S are S_d^k -visible. The cardinality of a largest d -distance k -mutual-visibility set in G is the d -distance k -mutual-visibility number of G , denoted by $\mu_d^k(G)$. One motivation for studying such a notion is that, two vertices $u, v \in V(G)$ are S_d^k -visible if there is, in a sense, a line of sight from u to v of length d that contains at most k obstacles from S . Such a situation may arise in practical applications in which viewers may have a certain degree of X-ray vision, so that they might be able to see through k or fewer obstacles but not through $k + 1$ obstacles.

Open Problem 7.7. For integers $d \geq 1$ and $k \geq 2$, determine $\mu_d^k(G)$ for various graphs.

Question 7.8. What results from the literature on mutual visibility in graphs can be generalized to the d -distance k -mutual-visibility?

Data availability

No data was used for the research described in the article.

References

- [1] Bijo S. Anand, Ullas Chandran S.V., Manoj Changat, Sandi Klavžar, Elias John Thomas, Characterization of general position sets and its applications to cographs and bipartite graphs, *Appl. Math. Comput.* 359 (2019) 84–89.
- [2] Peter Brass, On point sets without k collinear points, in: *Discrete Geometry*, in: *Monogr. Textbooks Pure Appl. Math.*, vol. 253, Dekker, New York, 2003, pp. 185–192.
- [3] Neal Bushaw, Brent Cody, Luke Freeman, Tobias Whitaker, The music and mathematics of maximally even sets, in: *BRIDGES Proceedings, 2024*, (available at <https://www.bridgesmathart.org/>).
- [4] Neal Bushaw, Brent Cody, Chris Leffler, Sets of vertices with extremal energy (preprint available at <https://arxiv.org/pdf/2407.18785>).
- [5] Martín Cera López, Pedro García-Vázquez, Juan Carlos Valenzuela-Tripodoro, Ismael G. Yero, The k -distance mutual-visibility problem in graphs, *Bull. Malays. Math. Sci. Soc.* 48 (1) (2025) Paper No. 25.
- [6] Ullas Chandran S.V., G. Jaya Parthasarathy, The geodesic irredundant sets in graphs, *Int. J. Math. Comb.* 4 (2016) 135.
- [7] John Clough, Jack Douthett, Maximally even sets, *J. Music Theory* 35 (1/2) (1991) 93–173.
- [8] John Clough, Gerald Myerson, Variety and multiplicity in diatonic systems, *J. Music Theory* 29 (2) (1985) 249–270.
- [9] Erik Demaine, Francisco Gomez-Martin, Henk. Meijer, David Rappaport, Perouz Taslakian, Godfried T. Toussaint, Terry Winograd, David R. Wood, The distance geometry of music, *Comput. Geom.* 42 (5) (2009) 429–454.
- [10] Gabriele Di Stefano, Mutual visibility in graphs, *Appl. Math. Comput.* 419 (2022) Paper No. 126850, 13.
- [11] Jack Douthett, Richard Krantz, Maximally even sets and configurations: common threads in mathematics, physics, and music, *J. Comb. Optim.* 14 (4) (2007) 385–410.
- [12] Jack Douthett, Richard Krantz, Dinner tables and concentric circles: a harmony of mathematics, music, and physics, *College Math. J.* 39 (3) (2008) 203–211.
- [13] Henry Ernest Dudeney, *Amusements in Mathematics*, Dover Publications, Inc., New York, 1959.
- [14] Paul Erdős, On some metric and combinatorial geometric problems, *Discrete Math.* 60 (1986) 147–153.
- [15] Paul Erdős, Some old and new problems in combinatorial geometry, in: *Applications of Discrete Mathematics* (Clemson, SC, 1986), SIAM, Philadelphia, PA, 1988, pp. 32–37.
- [16] Achim Flammenkamp, Progress in the no-three-in-line problem, *J. Combin. Theory Ser. A* 60 (2) (1992) 305–311.
- [17] Achim Flammenkamp, Progress in the no-three-in-line problem. II, *J. Combin. Theory Ser. A* 81 (1) (1998) 108–113.
- [18] Modjtaba Ghorbani, Hamid Reza Maimani, Mostafa Momeni, Farhad Rahimi Mahid, Sandi Klavžar, Gregor Rus, The general position problem on Kneser graphs and on some graph operations, *Discuss. Math. Graph Theory* 41 (4) (2021) 1199–1213.
- [19] R.R. Hall, T.H. Jackson, A. Sudbery, K. Wild, Some advances in the no-three-in-line problem, *J. Combin. Theory Ser. A* 18 (1975) 336–341.
- [20] Vesna Iršič, Sandi Klavžar, Gregor Rus, James Tuite, General position polynomials, *Results Math.* 79 (3) (2024) Paper No. 110, 16.
- [21] Dan Ismailescu, Restricted point configurations with many collinear k -tuples, in: *Discrete and Computational Geometry and Graph Drawing* (Columbia, SC, 2001), vol. 28, 2002, pp. 571–575.
- [22] Sandi Klavžar, Douglas F. Rall, Ismael G. Yero, General d -position sets, *Ars Math. Contemp.* 21 (1) (2021) Paper No. 3, 12.
- [23] Sandi Klavžar, Ismael G. Yero, The general position problem and strong resolving graphs, *Open Math.* 17 (1) (2019) 1126–1135.
- [24] Sandi Klavžar, Balázs Patkós, Gregor Rus, Ismael G. Yero, On general position sets in Cartesian products, *Results Math.* 76 (3) (2021) Paper No. 123, 21.
- [25] Sandi Klavžar, Gregor Rus, The general position number of integer lattices, *Appl. Math. Comput.* 390 (2021) Paper No. 125664, 4.
- [26] Danilo Korže, Aleksander Vesel, General position sets in two families of Cartesian product graphs, *Mediterr. J. Math.* 20 (4) (2023) Paper No. 203, 12.
- [27] Hanno Lefmann, Extensions of the no-three-in-line problem (preprint available at https://www.tu-chemnitz.de/informatik/ThIS/downloads/publications/lefmann_no_three_submitted.pdf).
- [28] Paul Manuel, Sandi Klavžar, A general position problem in graph theory, *Bull. Aust. Math. Soc.* 98 (2) (2018) 177–187.
- [29] Paul Manuel, Sandi Klavžar, The graph theory general position problem on some interconnection networks, *Fund. Inform.* 163 (4) (2018) 339–350.
- [30] Michael S. Payne, David R. Wood, On the general position subset selection problem, *SIAM J. Discrete Math.* 27 (4) (2013) 1727–1733.
- [31] Aleksander Misiak, Zofia Stepień, Alicja Szymaszkiewicz, Lucjan Szymaszkiewicz, Maciej Zwierzchowski, A note on the no-three-in-line problem on a torus, *Discrete Math.* 339 (1) (2016) 217–221.
- [32] Richard K. Guy, Patrick A. Kelly, The no-three-in-line problem, *Canad. Math. Bull.* 11 (1968) 527–531.
- [33] Zehui Shao, Aleksander. Vesel, Jin Xu, The k -distance independence number and 2-distance chromatic number of Cartesian products of cycles, *Bull. Malays. Math. Sci. Soc.* 41 (3) (2018) 1377–1391.
- [34] József Solymosi, Miloš Stojaković, Many collinear k -tuples with no $k + 1$ collinear points, *Discrete Comput. Geom.* 50 (3) (2013) 811–820.
- [35] Perouz Taslakian, *Musical Rhythms in the Euclidean Plane* (Ph.D. thesis), McGill University, Montreal, Quebec H3A 0G4, Canada, 2008.
- [36] Jing Tian, Kexiang Xu, Sandi Klavžar, The general position number of the Cartesian product of two trees, *Bull. Aust. Math. Soc.* 104 (1) (2021) 1–10.
- [37] Godfried Toussaint, The geometry of musical rhythm, in: *Discrete and Computational Geometry*, in: *Lecture Notes in Comput. Sci.*, vol. 3742, Springer, Berlin, 2005, pp. 198–212.